7

Trees

Chapter Learning Outcomes

CLO

7

- Know and understand the meaning of tree as a data structure.
- Understand some terminologies for rooted trees (parent, child, siblings, ancestors, descendants).
- Distinguish between internal and external nods.
- Understand the three methods for tree traversal.
- Create the expression tree for a given expression.
- Understand the three methods for traversing expression trees.

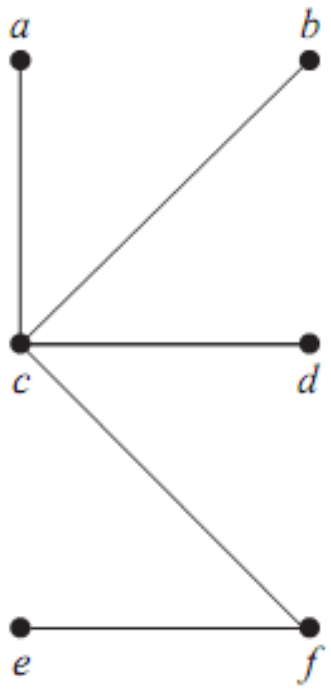
المعيدي

Trees

A ***tree*** is a connected undirected graph with no simple circuits.

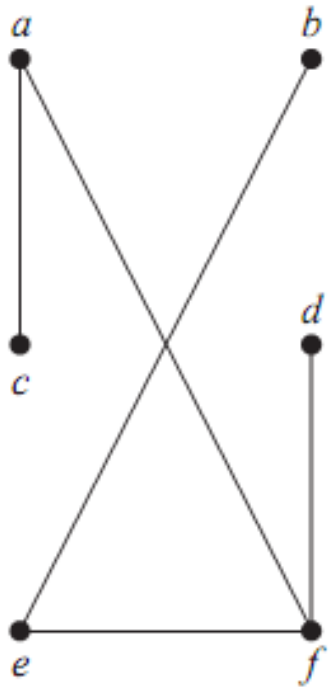
EXAMPLE

Determine if the following graph is a tree?



G_1

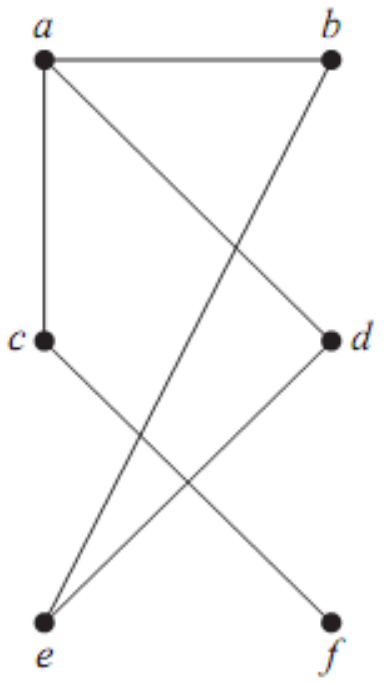
EXAMPLE Determine if the following graph is a tree?



G_2

د. حسن المعيدي

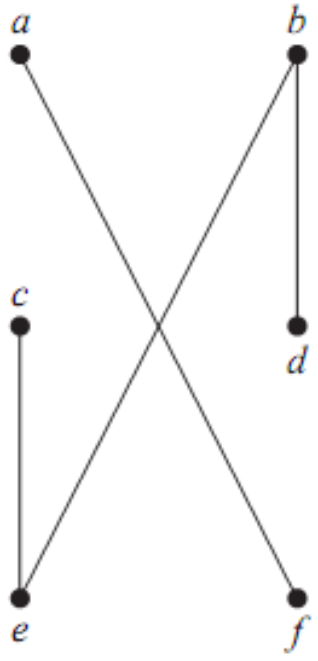
EXAMPLE Determine if the following graph is a tree?



G_3

د. حسن المعيدي

EXAMPLE Determine if the following graph is a tree?

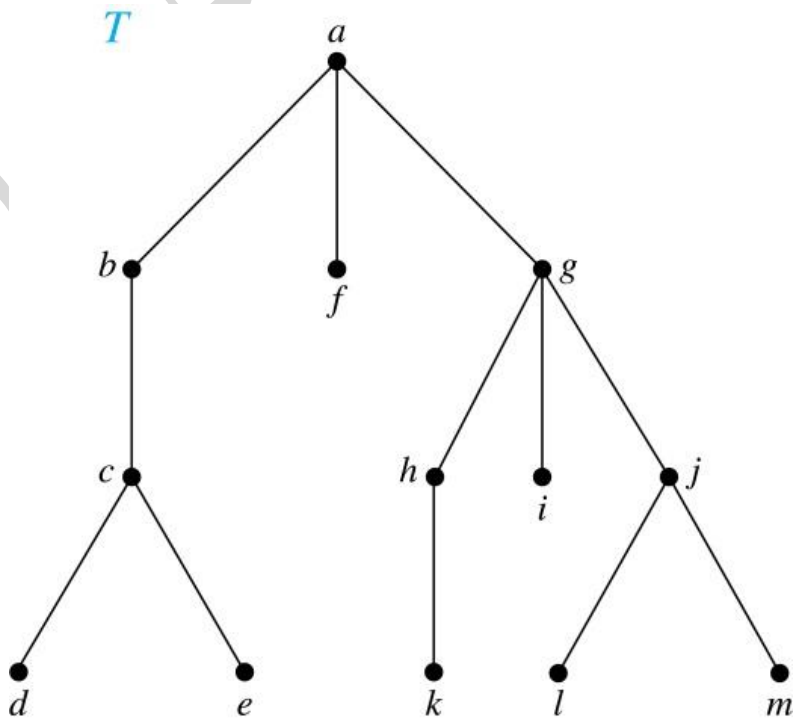


G_4

د. حسن المعيدي

Rooted Tree

A ***rooted tree*** is a tree in which one vertex has been used as the *root* and every edge is directed away from the root.



Terminology for Rooted Trees

1- Find the parent of *c.*

2- Find the children of *g.*

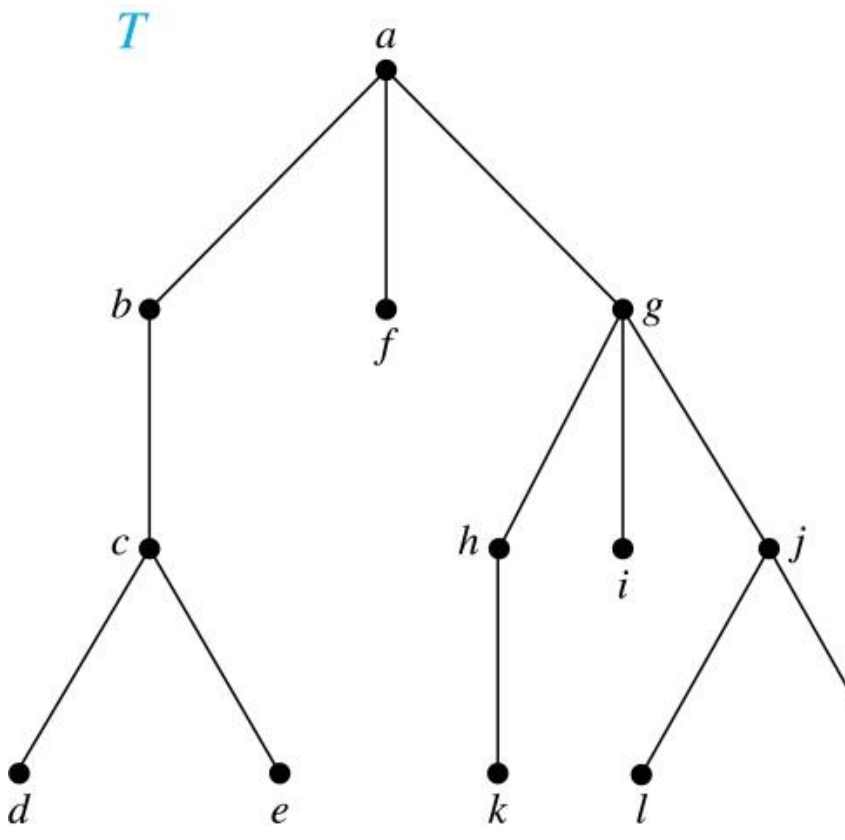
3- Find the siblings of h.

4- Find the ancestors of e.

5- Find the descendants of b.

6- Find all internal nodes.

7- Find all leaves (external nodes).



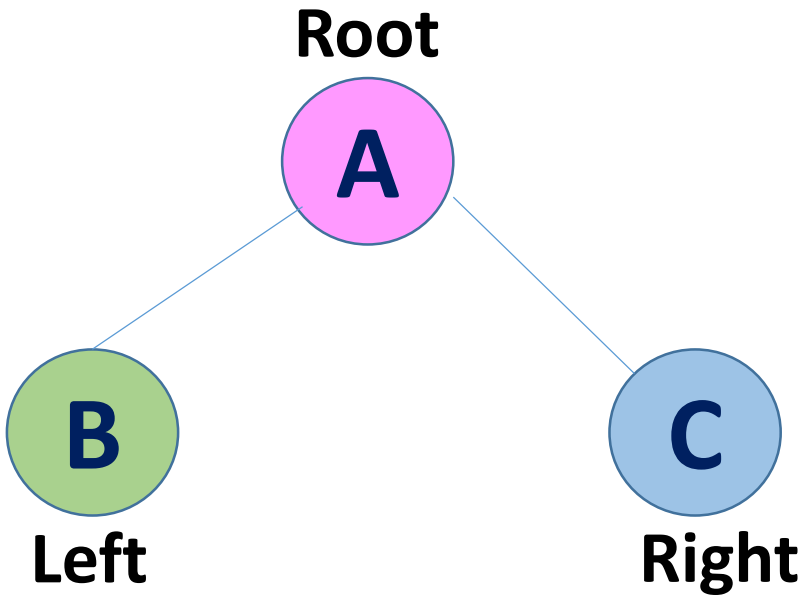
د. حسن المعيدي

Tree Traversal

Procedures for visiting every vertex of an ordered tree are called *traversals*.

There are three different traversals:

- *Preorder traversal.*
- *Inorder traversal.*
- *Postorder traversal.*

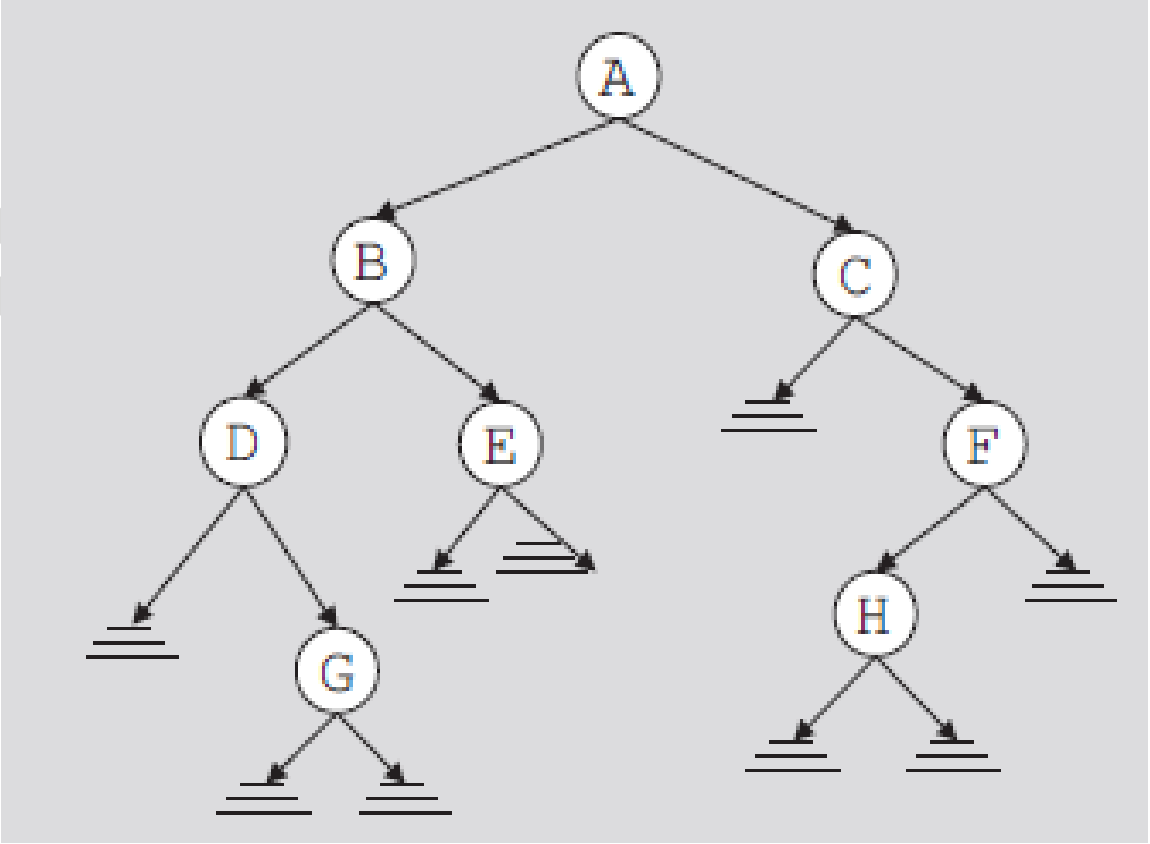


Preorder traversal	A	B	C
Inorder traversal	B	A	C
Postorder traversal	B	C	A

د. حسن المعيدي

Inorder Traversal ($_{Left}$-Root-$_{Right}$)

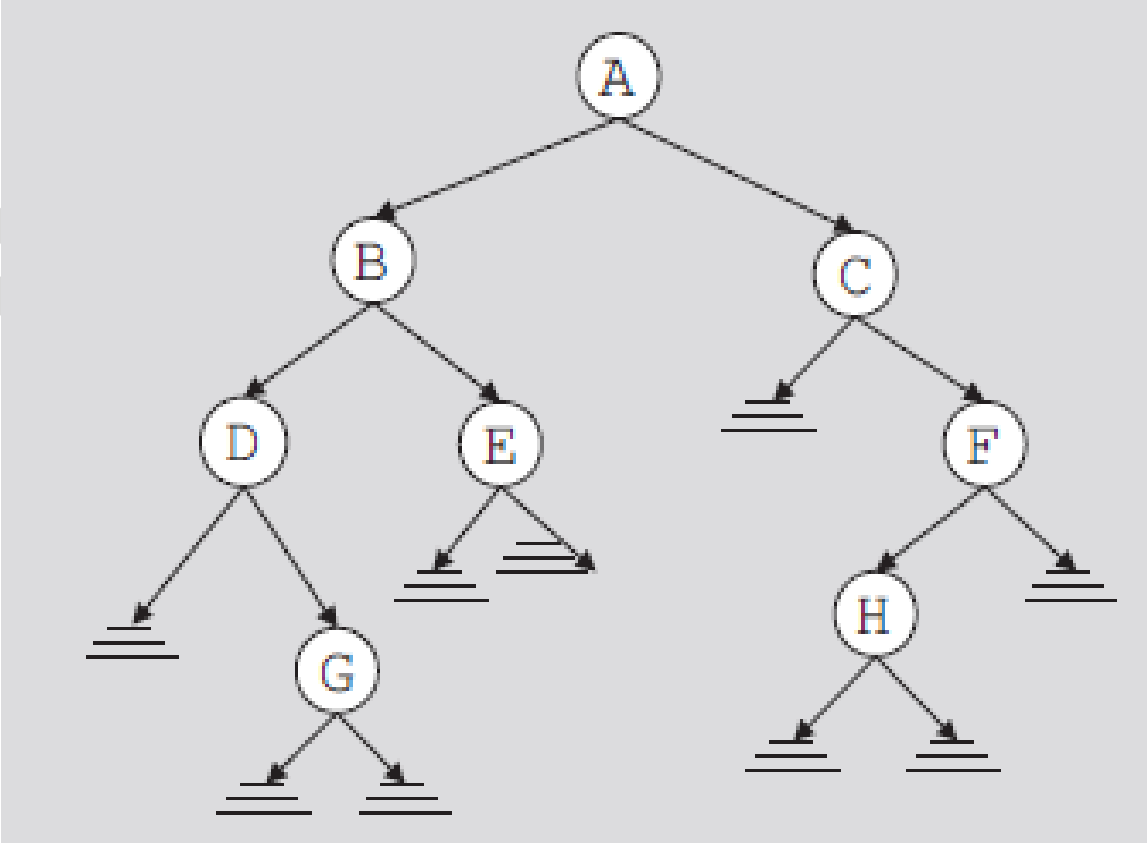
- Traverse the left subtree
- Visit the node (Root)
- Traverse the right subtree



حسن المعيدي

Preorder Traversal (Root-Left-Right)

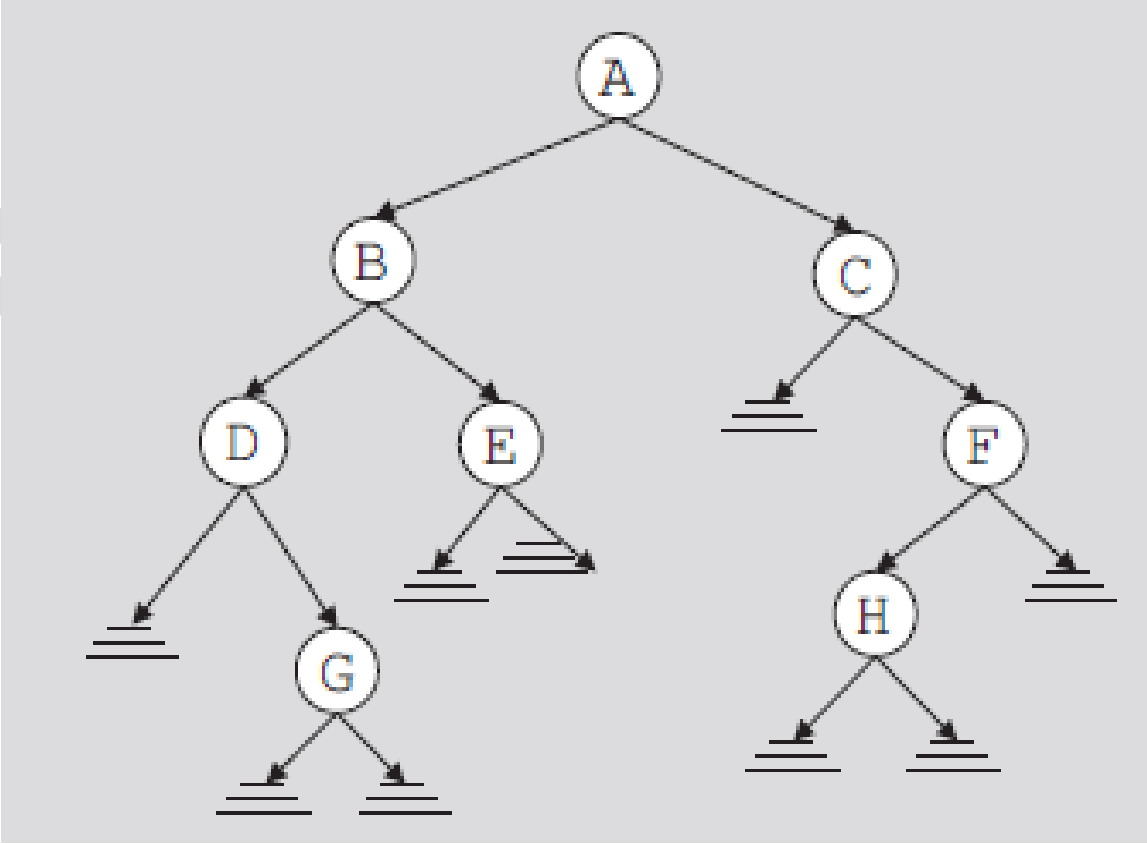
- Visit the node (Root)
- Traverse the left subtree
- Traverse the right subtree



حسن المعيدي

Postorder Traversal (Left-Right-Root)

- Traverse the left subtree
- Traverse the right subtree
- Visit the node (Root)



حسن المعيدي

Example

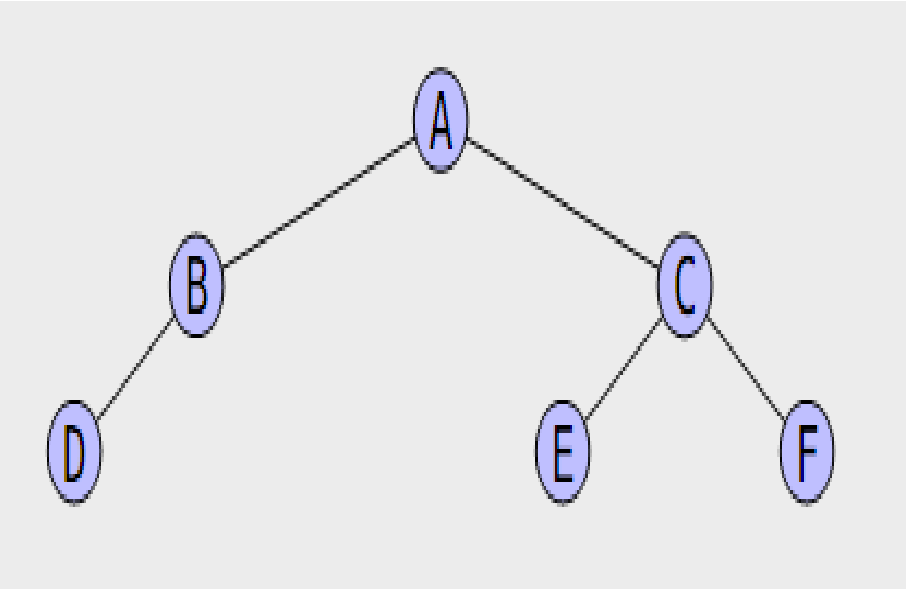
EXAMPLE

In what order are the nodes visited by an inorder, postrorder and preorder traversal of this tree?

Inorder

Preorder

Postorder



د. حسن المعيدي

Expression Trees

Binary tree for an arithmetic expression

- **Internal nodes : operators**
- **Leaves : operands**

EXAMPLE

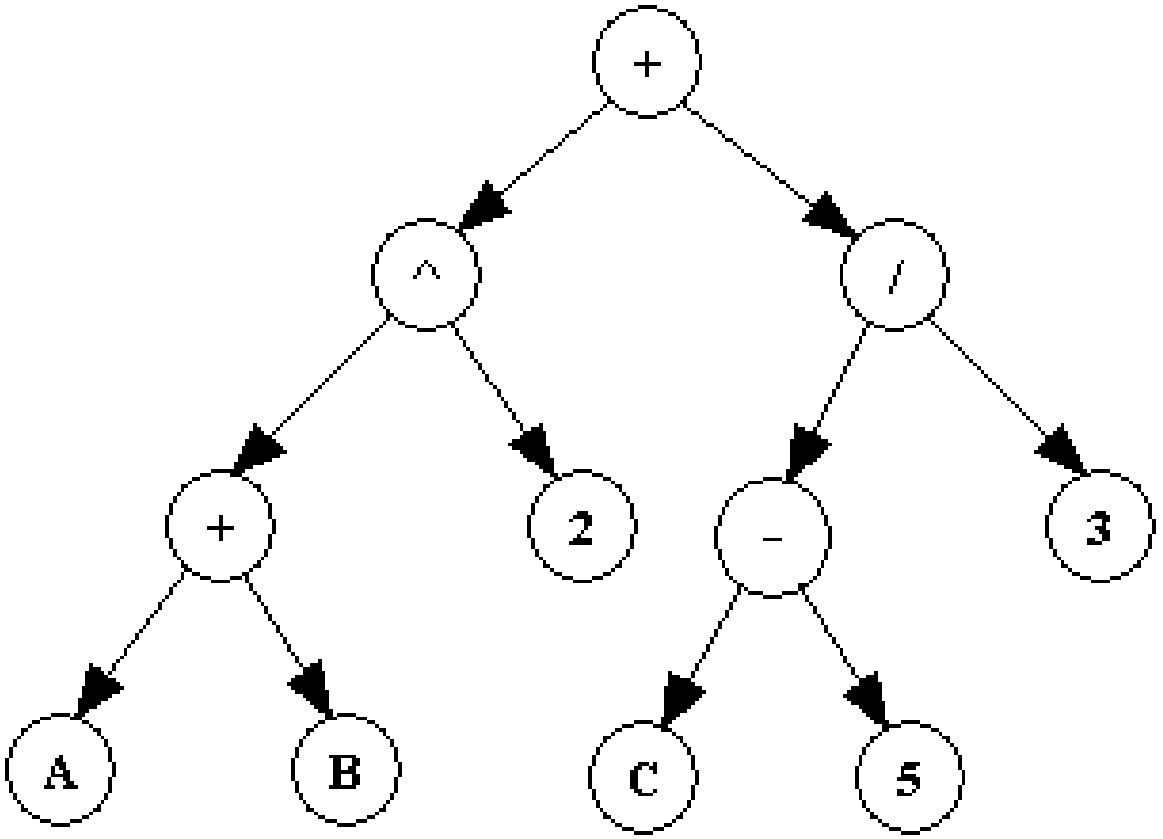
Example: arithmetic expression tree for the expression:
$2 \times (5 - 1) + (3 \times 2)$

د. حسن المعيدي

Expression Trees

EXAMPLE

Create the expression tree of $(A + B)^2 + (C - 5) / 3$



Traversing Expression Trees

Infix	Inorder	L-R-r
Prefix	Preorder	R-L-r
Postfix	Postorder	L-r-R

د. حسن المعيدي

Advantages

- The need for parentheses is eliminated.
- The precedence rule is no longer relevant.

د. حسن المعيدي

Traversing Expression Trees

EXAMPLE

What is the value of following postfix expression:

Postfix : 3 4 5 * +

د. حسن المعيدي

Traversing Expression Trees

EXAMPLE

What is the value of following postfix expression:

Postfix : 7 2 + 3 *

د. حسن المعيدي

Quiz

EXAMPLE

What is the value of following postfix expressions:

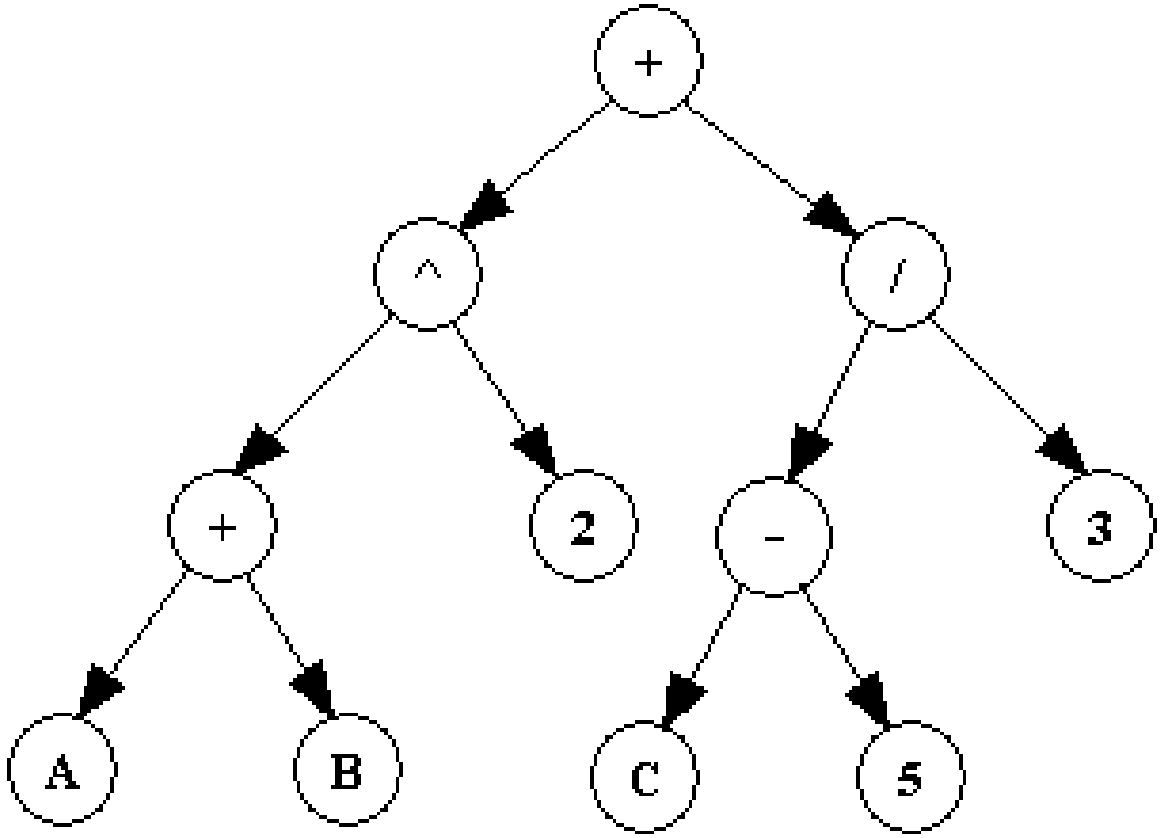
- 5 2 1 – – 3 1 4 ++ ∗
- 9 3 / 5 + 7 2 – ∗
- 3 2 ∗ 2 ^ 5 3 – 8 4 / ∗ –

Expression Trees

EXAMPLE

Write the infix notation for the following tree.

L-R-r



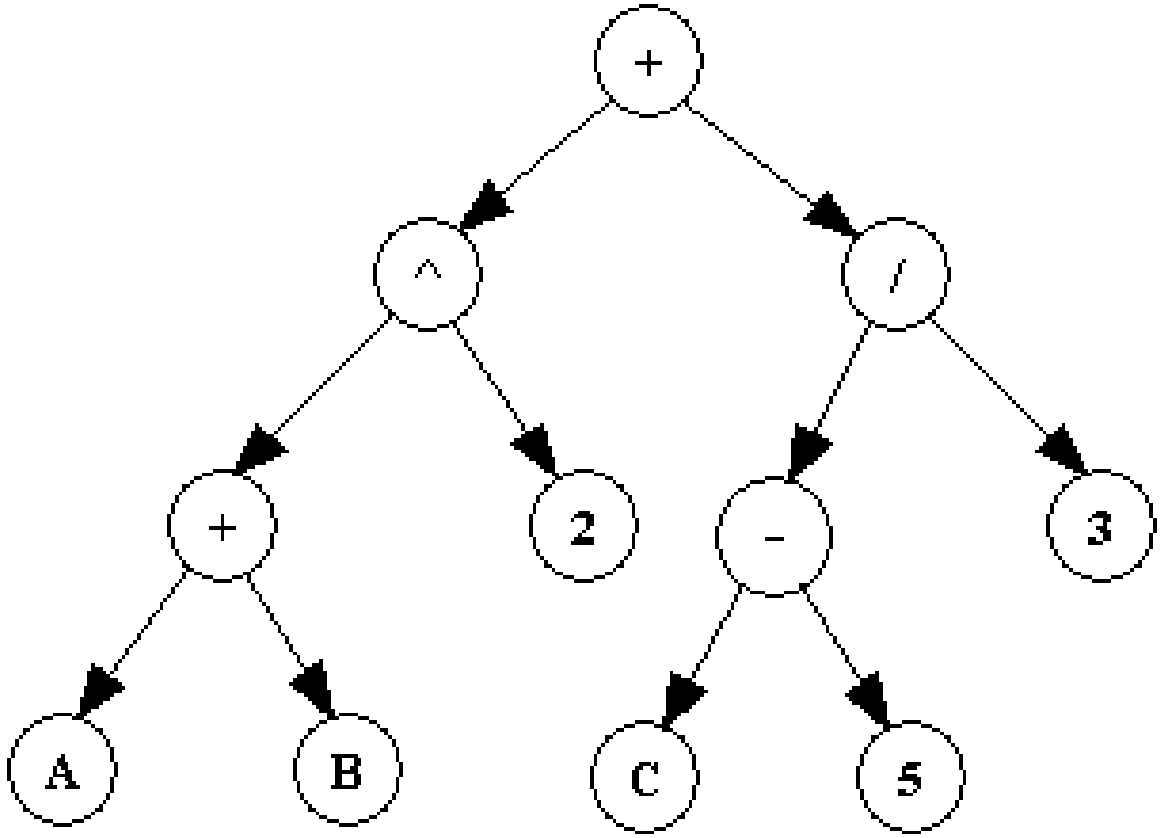
حسن المعيدي

Expression Trees

EXAMPLE

Write the prefix notation for the following tree.

R-L-r



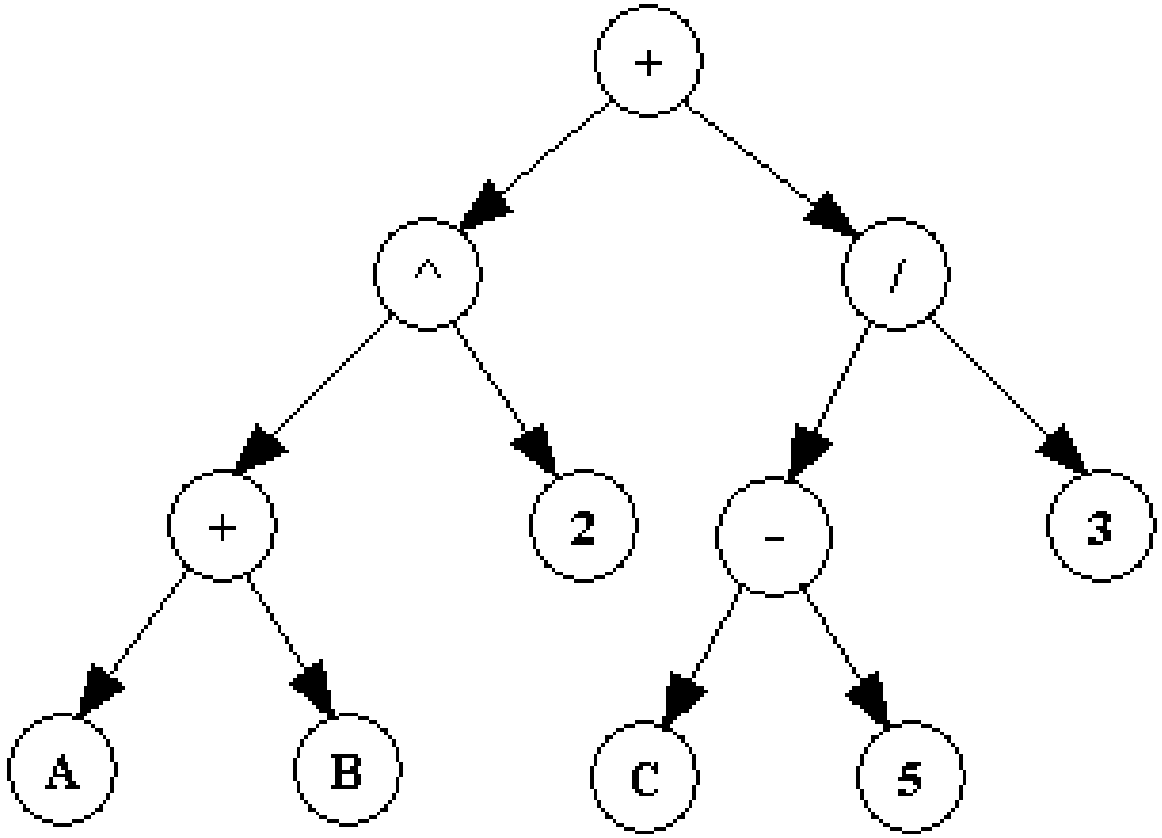
حسن المعيدي

Expression Trees

EXAMPLE

Write the postfix notation for the following tree.

L-r-R



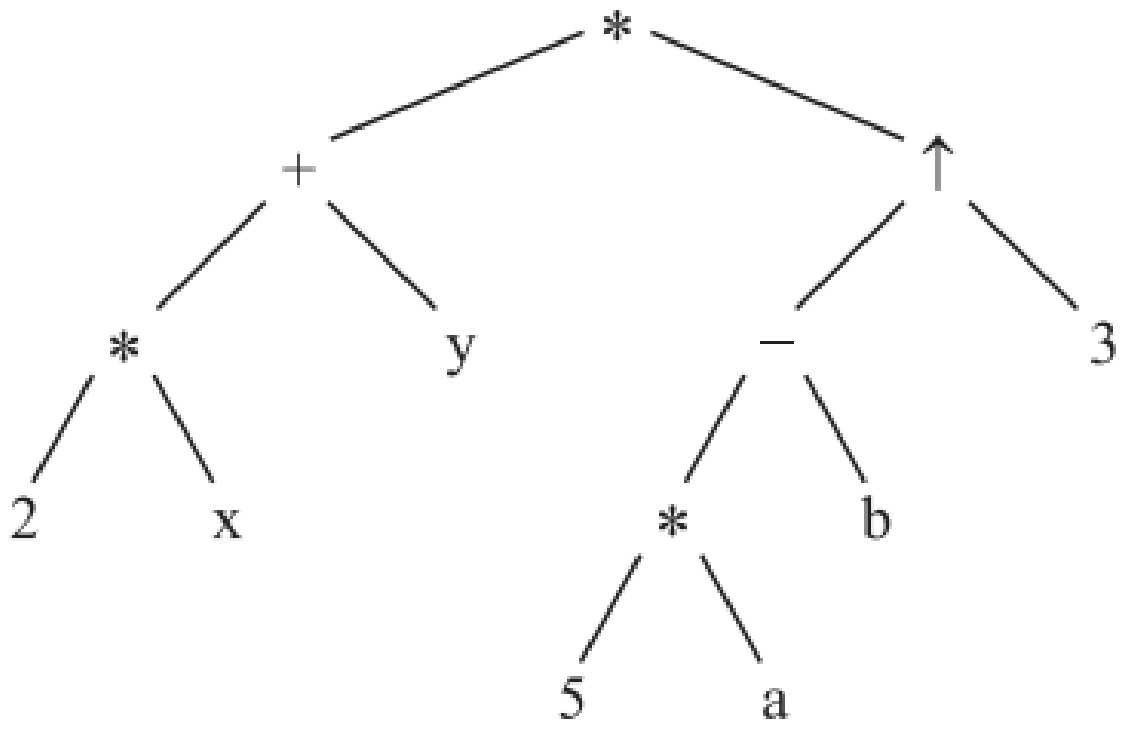
حسن المعيدي

Expression Trees

EXAMPLE

Write the infix notation for the following tree.

L-R-r



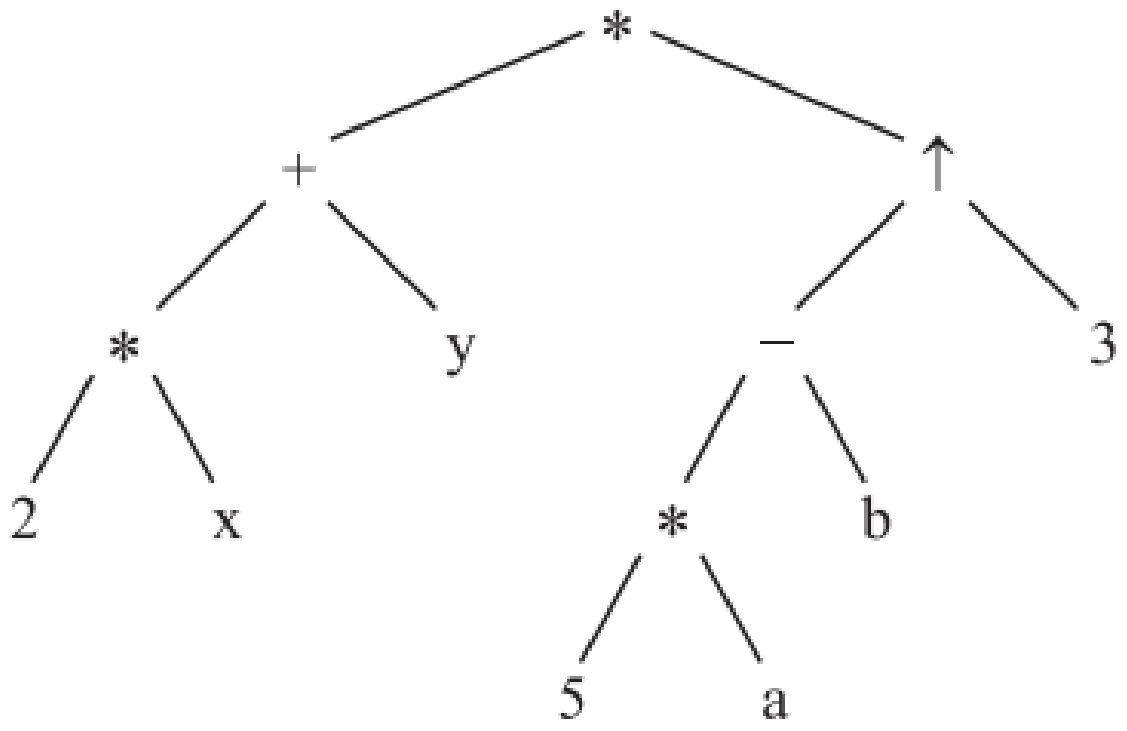
د. حسن المعيدي

Expression Trees

EXAMPLE

Write the prefix notation for the following tree.

R-L-r

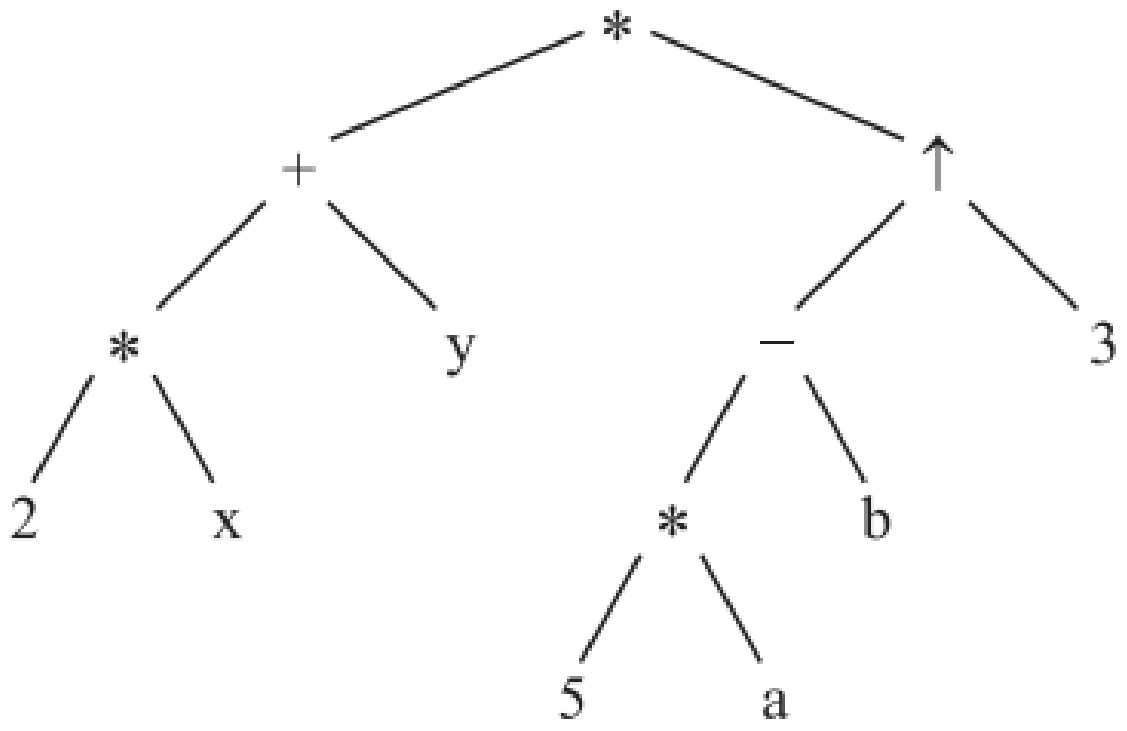


Expression Trees

EXAMPLE

Write the postfix notation for the following tree.

L-r-R



حسن المعيدي

Check - Chapter Learning Outcomes

CLO

7

- Know and understand the meaning of tree as a data structure.
- Understand some terminologies for rooted trees (parent, child, siblings, ancestors, descendants).
- Distinguish between internal and external nods.
- Understand the three methods for tree traversal.
- Create the expression tree for a given expression.
- Understand the three methods for traversing expression trees.

المعيدي

End of Chapter

7